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The essentials

Introduction and Motivation

LDA (Linear Discriminant Analysis) is a **supervised latent analysis** method that makes it possible to find optimal latent factors for discrimination between classes.

The fundamental problem

Imagine that you work in a postal sorting center. You have letters with 100 different measurements (weight, size, color, texture…). You must sort them automatically by type (letter, parcel, magazine). How can you find **the dimension or dimensions that best separate the categories**?

This is exactly what LDA solves: finding **the axes that maximize separation between classes** while minimizing dispersion within each class.

Fundamental difference with PCA:

PCA looks at **all** the data and searches for directions of maximum variation. LDA first looks at **the labels** (the classes) and searches for the directions that best separate these classes.

Warning: This LDA should not be confused with **Latent Dirichlet Allocation** (used in NLP).

Supervised learning

Let data $X \subset \Omega \subseteq \mathbb{R}^D$ be associated with categories (class labels) $Y = [[M]] \subset \mathbb{N}$.

Supervised learning consists of finding (learning) the parameters θ of a learner (function) ϕ_θ in order to minimize the loss $L_{X \times Y}(\theta)$ incurred when predicting class labels.

$$\phi_\theta : \Omega \to Y$$
$$x_i \mapsto \phi_\theta(x_i) = \tilde{y}_i$$

Geometric intuition: Separating colored point clouds

Imagine points of two colors (red and blue) scattered on a plane. You can only look along a single direction (like a flashlight that illuminates a line).

Question: How should the flashlight be oriented so that, on this line, the red and blue points are **as separated as possible**?

Essential idea: LDA looks for the **dividing line**, like a referee drawing a line between two teams. It finds the direction that best separates the groups, while ensuring that each team remains clustered.

We seek a **direction** $a \in \mathbb{R}^D$ such that the projection of the data onto this direction $\hat{X} = \mathrm{Proj}_a(X)$ exhibits **optimal properties for discrimination** between classes.

The two competing objectives

Objective 1: Move the centers apart (Between-class)

"Push the means as far apart as possible"

If the class projections are **well separated**, then they can be easily discriminated.

Objective: Search for the direction a where inter-class discrimination is **maximal**.

> **Mathematical details: Between-class formulation**
>
> The direction a must be **collinear** with the line $\mu_1 - \mu_2$ that connects the centers of mass of the classes, because:
>
> $$|\hat{\mu}_1 - \hat{\mu}_2|^2 = \left| \frac{a^T \mu_1}{|a|^2} a - \frac{a^T \mu_2}{|a|^2} a \right|^2 = \frac{1}{|a|^2} (a^T(\mu_1 - \mu_2))^2$$
>
> This expression is **maximal** when $a = \lambda(\mu_1 - \mu_2)$.
> **Property:** The center of mass of the projected data is the projection of the center of mass of the original data.

Objective 2: Group each class (Within-class)

"Keep each class compact"

We also seek to **minimize the variance** of each class when projected individually.

For each class k, the normalized variance of the projection is:

$$\hat{\sigma} * k^2 = \frac{1}{N_k} \sum *y_i = k(\hat{x}_i - \hat{\mu}_k)^T (\hat{x}_i - \hat{\mu}_k)$$

The dilemma: Sometimes, pulling the centers apart separates them, but it can also stretch the classes. The right compromise must be found.

Fisher criterion: The mathematical compromise

Formulation for 2 classes

The **Fisher criterion** maximizes the ratio:

$$\arg\max_{a} \frac{|\hat{\mu}_1 - \hat{\mu}_2|^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}$$

Interpretation:

- **Numerator:** Distance between projected means (between-class) → to **maximize**
- **Denominator:** Sum of within-class variances → to **minimize**

Mnemonic trick:

- **B** as in **Between** = **B** for "Separate well" (the centers)
- **W** as in **Within** = "Group together" (each class)

Generalization to M classes

Mathematical details: Covariance matrices

Between-class covariance matrix:
Let $B = [\mu_1 - \mu, \dots, \mu_M - \mu]$ be the matrix of centered class means, where:

- $\mu = \frac{1}{N}\sum_i^N x_i$ is the global mean
- $N = \sum_k N_k$ is the total number of points

$$\Sigma_b = \frac{1}{M} B B^T$$

Criterion to maximize:

$$\frac{1}{M}\sum_k (\hat{\mu}_k - \hat{\mu})^T(\hat{\mu}_k - \hat{\mu}) = \frac{1}{|a|^2} a^T \Sigma_b a$$

Within-class covariance matrix:
Let $A_k = [x_1 - \mu_k, \dots, x_{N_k} - \mu_k]$ with $y_i = k$, the matrix of centered data for class k.

$$\Sigma_w = \sum_k \frac{1}{N_k} A_k A_k^T$$

Criterion to minimize:

$$\sum_k \frac{1}{N_k} \sum_{y_i=k} (\hat{x}_i - \hat{\mu}_k)^T (\hat{x}_i - \hat{\mu}_k) = \frac{1}{a^T a} a^T \Sigma_w a$$

The **generalized Fisher criterion** combines the between-class and within-class criteria:

$$\arg\max_a, J(a) = \arg\max_a \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

where:

- Σ_b = "between-class" covariance matrix → measures separation of centers
- Σ_w = "within-class" covariance matrix → measures internal dispersion

---

Solution: Eigenvalue problem

Solution by differentiation

Mathematical details: Derivation

To find the maximum, we compute the derivative:

$$\frac{\partial J(a)}{\partial a} = \frac{\Sigma_b a (a^T \Sigma_w a) - \Sigma_w a (a^T \Sigma_b a)}{(a^T \Sigma_w a)^2} = 0$$

Resulting equation:

$$\Sigma_b a = J(a) \Sigma_w a$$

Which can be rewritten as:

$$\Sigma_w^{-1} \Sigma_b a = J(a) a$$

Solution: a is a solution of the **generalized eigenvalue system**.

The optimal direction a is the **first eigenvector** of $\Sigma_w^{-1} \Sigma_b$ (with eigenvalue $\lambda_1 = J(a)$).

Dimensional limitation

Why only M-1 dimensions?

The eigenvectors corresponding to the **largest eigenvalues** λ_i are the most discriminative dimensions:

$$a_1, a_2, \dots, a_p \quad \text{with} \quad \lambda_1 > \lambda_2 > \dots > \lambda_p$$

Fundamental property: M classes can be discriminated in a subspace of dimension **at most** $(M-1)$.

Consequence: There are only $M-1$ **non-zero** eigenvalues.

Analogy: With 3 points (3 classes), you can at most place them in a plane (2D). With 2 points (2 classes), you can at most place them on a line (1D).

> **i** Mathematical details: Case M = 2
>
> For two classes, one can show that:
>
> $$BB^T = (\mu_1 - \mu)(\mu_1 - \mu)^T + (\mu_2 - \mu)(\mu_2 - \mu)^T \propto (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$
>
> **Implications:**
>
> - $BB^T a$ is a vector in the direction $(\mu_1 - \mu_2)$
> - The solution is $a = \lambda \Sigma_w^{-1}(\mu_1 - \mu_2)$

Pitfall to avoid: If you keep more than M-1 dimensions, the additional directions will have a zero eigenvalue → not discriminative.

Practical application of LDA

Complete LDA algorithm

Step 1: Prepare the data

- Check that each class has enough examples (at least D+1 to avoid singular Σ_w)
- Optional: standardize if the scales are very different

Step 2: Compute the key matrices

- Σ_w (within): average of class covariances
- Σ_b (between): covariance of class centers

Step 3: Solve the eigenvalue problem

Compute the eigenvectors of $\Sigma_w^{-1}\Sigma_b$

Step 4: Keep the first M-1 directions

Sort by decreasing eigenvalues, keep the most important ones

Step 5: Project and classify

- Project the data onto the discriminant subspace
- Classify (for example using a nearest-neighbor rule)

LDA as a classifier: The Gaussian hypothesis

Probabilistic modeling

For a new data point j:

- x_j is **known**
- y_j is **unknown**

Question: To which class k does point j belong?

We define the class random variable C and use **Bayes' rule**:

$$\mathbb{P}(C = k|x_j) = \frac{\mathbb{P}(x_j|C = k)\mathbb{P}(C = k)}{\mathbb{P}(x_j)}$$

Components:

- $\mathbb{P}(x_j|C = k)$: **likelihood**
- $\mathbb{P}(C = k)$: **prior**
- $\mathbb{P}(x_j)$: **evidence** (normalization)

Gaussian approximation

LDA assumes that **each class follows a Gaussian distribution**:

$$\mathbb{P}(x|C=k) \sim \mathcal{N}(\mu_k, \Sigma_k)$$

with:

- Different means for each class
- **The same covariance matrix** for all classes (crucial!)

> **i** Mathematical details: Maximum likelihood
>
> **Approximation:**
>
> $$\mathbb{P}(x|C=k) \propto \exp\left(-(x-\mu_k)^T \Sigma_k^{-1} (x-\mu_k)\right)$$
>
> **Uniform prior:** $\mathbb{P}(C=k) = \frac{1}{M}$ (all classes equally likely)
> **Simplification:** The evidence $\mathbb{P}(x_j)$ is **ignored** because it is constant for all classes (normalization).

Decision rule:

$$\delta(x) = \arg\max_k, \mathbb{P}(x|C=k)$$

Interpretation: x is assigned to the class k that maximizes the likelihood.

Why is the boundary linear?

With this assumption, the boundary between two classes is **linear** (hence the "L" in LDA).

Why? The equal-density contours of two Gaussians with the same covariance are hyperplanes.

Optimality of LDA

Optimality conditions

LDA is optimal when the M classes each follow a **Gaussian distribution** with the same covariance matrix.

Justification:

The discrimination criteria are based on the covariance matrices Σ_w and Σ_b, which fully characterize Gaussian distributions.

When LDA works well (and less well)

Ideal conditions

- Classes are **approximately Gaussian**
- Class covariances are **similar**
- Enough samples per class (at least 10× the number of dimensions)
- **Linear** relationships between variables

Problematic situations

- **Very different covariances** → use QDA (Quadratic Discriminant Analysis)
- **Non-linear relationships** → Kernel LDA or non-linear methods
- **Few samples** relative to dimensions → regularize or reduce dimensionality first
- **Non-Gaussian classes** → other classifiers (random forests, SVM…)

LDA vs QDA: The linear vs quadratic dilemma

The crucial difference

- **LDA:** Assumes **the same covariance** for all classes → linear boundaries
- **QDA:** Assumes **different covariances** → quadratic boundaries

Practical rule: If you have little data, LDA is more robust (fewer parameters to estimate).

LDA vs PCA comparison

Fundamental differences

PCA (unsupervised):

- Maximizes **global variance**
- Ignores class labels
- Finds directions of maximum variance
- Based on Σ (total covariance)

LDA (supervised):

- Maximizes **class separation**
- Uses class labels
- Finds discriminative directions
- Based on $\Sigma_w^{-1}\Sigma_b$ (covariance ratio)

Complementarity

In practice: PCA and LDA can be combined:

- PCA for initial dimensionality reduction (noise removal)
- LDA to maximize discrimination

Concrete example: Face recognition with Fisherfaces (LDA) as opposed to Eigenfaces (PCA).

Condensed revision sheet

In 30 seconds:

- LDA = supervised dimensionality reduction (uses labels)
- Maximizes (distance between centers) / (dispersion within classes)
- Solution = eigenvectors of $\Sigma_w^{-1}\Sigma_b$
- At most M-1 discriminative dimensions
- Assumes: Gaussian classes, identical covariances
- Useful for: classification, supervised visualization
- Alternative: QDA if covariances differ

Mathematical concepts to master

To fully understand LDA:
Linear algebra:

- Eigenvectors and eigenvalues
- Generalized eigenvalue system
- Covariance matrices
- Orthogonal projection

Statistics:

- Mean and variance
- Covariance and correlation
- Multivariate normal distribution
- Bayes' rule

Optimization:

- Matrix function differentiation
- Constrained optimization
- Rayleigh quotient method